9 The Sample Mean and Variance Together

Imagine checking a machine that cuts cable into pieces. You would probably ask two questions. Are the pieces centered near the intended length? And how much do their lengths vary?

The sample mean answers the first question, while the sample variance helps answer the second. Both numbers come from the same observations. That raises a new question: if a sample has an unusually large mean, does it also tend to have an unusually large variance?

For independent observations from a normal population, the answer is no. The sample mean and sample variance are independent. This is a special feature of the normal model, and it is the key connection between this chapter and the t distribution in Chapter 10.

9.1 Two estimates from one sample

Suppose we observe an independent random sample

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad n \geq 2,$$

where both the population mean μ and population variance $\sigma^2 > 0$ are unknown. Unlike the known-mean setting in Chapter 8, we must estimate the center before calculating deviations from it.

The MLE of the mean is the sample mean:

$$\hat{\mu} = \overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

To estimate spread, we start by adding the squared distances from this sample mean. There are two common ways to turn that sum into a variance estimate.

> **Note**
>
> Let Q denote the sum of squared deviations from the sample mean:

$$Q = \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

The normal-model MLE of the variance is $\hat{\sigma}^2 = Q/n$. The usual sample variance is $S^2 = Q/(n-1)$. The letter S denotes the sample standard deviation, so $S = \sqrt{S^2}$. Lowercase s^2 and s denote the variance and standard deviation calculated from observed data. The population quantities are σ^2 and σ. Neither S^2 nor $\hat{\sigma}^2$ should be confused with the population variance itself.
Here Q has squared measurement units. It will become a chi-square variable only after we divide it by σ^2.

Here we will focus on the MLE, which divides by n. We will keep both versions visible because the usual sample variance, which divides by $n-1$, appears in the t distribution.

The estimates are related by

$$S^2 = \frac{n}{n-1}\hat{\sigma}^2.$$

9.1.1 Example: four thickness measurements

Suppose four prototype panels have measured thicknesses of 9, 10, 12, and 13 millimeters. Their sample mean is

$$\overline{x} = \frac{9 + 10 + 12 + 13}{4} = 11.$$

The deviations from the sample mean are $-2, -1, 1, 2$, so the sum of squared deviations is

$$(-2)^2 + (-1)^2 + 1^2 + 2^2 = 10.$$

The two variance estimates are therefore

$$\hat{\sigma}^2 = \frac{10}{4} = 2.5, \qquad s^2 = \frac{10}{3} \approx 3.333.$$

Both have units of square millimeters. The sample standard deviation is $s = \sqrt{10/3} \approx 1.826$ millimeters.

```
thickness <- c(9, 10, 12, 13)
mean(thickness)
```

```
[1] 11
```

```
mean((thickness - mean(thickness))^2)
```

```
[1] 2.5
```

```
var(thickness)
```

```
[1] 3.333333
```

```
sd(thickness)
```

```
[1] 1.825742
```

In R, `var()` uses the divisor $n-1$, and `sd()` takes the square root of that result. To calculate the MLE directly, the code averages the squared deviations using `mean()`.

9.2 The three results we need

Under independent normal sampling, the following statements hold exactly:

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right),$$

$$\frac{Q}{\sigma^2} = \frac{n\hat{\sigma}^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

and the sample mean is independent of either version of the sample variance.

The first statement describes where sample means tend to fall. The second describes the sample's spread. The third says that, under the normal model, knowing one of these summaries does not change the sampling distribution of the other.

These are exact normal-sampling results, even for a small sample. They do not require a large-sample approximation.

9.3 Why the sample mean is normal

A sum of independent normal random variables is normal. Its mean is the sum of their means, and its variance is the sum of their variances. Thus,

$$\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2).$$

Dividing the sum by n divides the mean by n and the variance by n^2. This gives

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

The fact that σ^2 is unknown does not change this sampling distribution. It changes what we can calculate from the data. The true standard error is still $\sigma/\sqrt{n}$, but we will need to estimate it when using the sample to make an inference.

9.4 Why estimating the mean costs one degree of freedom

In Chapter 8, we centered observations at a known population mean. There were n independent standardized deviations, and their squared sum had n degrees of freedom.

Here we center at the sample mean. The resulting deviations must add to zero.

> **Note**
>
> $R_i = X_i - \overline{X}$ is called the residual for observation i. It is the difference between that observation and the fitted center of the sample.
> The identity $\sum_{i=1}^{n} R_i = 0$ means that the residuals balance exactly. Once the first $n-1$ residuals are known, the last must be
>
> $$R_n = -\sum_{i=1}^{n-1} R_i.$$

For the four panels, the residuals were $-2, -1, 1, 2$. If we had only been told the first three, we could already work out the fourth. It must equal 2 to make the total zero.

This explains the loss of one degree of freedom. There are n residuals, but only $n-1$ can vary freely once their sum is fixed. The residuals are not independent, so we cannot simply treat their squared values as n independent chi-square contributions.

9.4.1 Connecting the residuals to a chi-square distribution

Standardize the original observations using the population mean and standard deviation:

$$Z_i = \frac{X_i - \mu}{\sigma}.$$

The Z_i are independent standard normal variables. Their sample mean is $\overline{Z} = (\overline{X} - \mu)/\sigma$, so

$$Z_i - \overline{Z} = \frac{X_i - \overline{X}}{\sigma}.$$

A result for standard normal samples states that

$$\sum_{i=1}^{n} (Z_i - \overline{Z})^2 \sim \chi^2_{n-1}.$$

Applying it here gives

$$\frac{Q}{\sigma^2} = \sum_{i=1}^{n} (Z_i - \overline{Z})^2 \sim \chi^2_{n-1}.$$

9.4.2 Seeing the result with just two observations

For two independent normal observations, the deviations from their sample mean are

$$X_1 - \overline{X} = \frac{X_1 - X_2}{2}, \qquad X_2 - \overline{X} = -\frac{X_1 - X_2}{2}.$$

There is only one distinct piece of information about spread: the difference between the observations. Their sum of squared deviations is

$$Q = \frac{(X_1 - X_2)^2}{2}.$$

The difference $X_1 - X_2$ is normal with mean 0 and variance $2\sigma^2$. Therefore,

$$\frac{Q}{\sigma^2} = \left(\frac{X_1 - X_2}{\sqrt{2}\sigma} \right)^2 \sim \chi^2_1.$$

The squared standard normal on the right makes the single remaining degree of freedom visible.

9.5 What the variance distributions tell us

Since a chi-square variable with $n-1$ degrees of freedom has mean $n-1$,

$$E[Q] = (n-1)\sigma^2.$$

Dividing by n gives

$$E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2.$$

The MLE underestimates the population variance on average. Its bias is $-\sigma^2/n$.

Dividing by $n-1$ corrects this bias:

$$E[S^2] = \sigma^2.$$

This is why the usual sample variance uses $n-1$. It is a correction for having estimated the center from the same data.

The chi-square variance also gives

$$\mathrm{Var}(Q) = 2(n-1)\sigma^4.$$

Consequently,

Estimator	Expected value	Sampling variance
$\hat{\sigma}^2 = Q/n$	$(n-1)\sigma^2/n$	$2(n-1)\sigma^4/n^2$
$S^2 = Q/(n-1)$	σ^2	$2\sigma^4/(n-1)$

The unbiased estimator has a slightly larger sampling variance because it multiplies the MLE by a constant greater than one. Unbiasedness and small sampling variance describe different properties, so neither one alone settles every comparison between estimators.

The gamma forms are useful too. Using the shape-rate convention from Chapter 8,

$$\widehat{\sigma}^2 \sim \text{Gamma}\left(\frac{n-1}{2}, \frac{n}{2\sigma^2}\right),$$

$$S^2 \sim \text{Gamma}\left(\frac{n-1}{2}, \frac{n-1}{2\sigma^2}\right).$$

The two distributions have the same shape parameter. Their different rate parameters reflect their different denominators.

9.6 Why the mean and variance are independent

It may seem surprising that two summaries of the same observations can be independent. A useful first step is to separate a change in location from a change in spread.

If we add 3 to every observation, the sample mean increases by 3. Every residual stays the same because the new sample mean also increases by 3. So the sample variance is unchanged. This shows that the summaries measure different features. By itself, however, it does not prove independence.

For the normal model, the key algebraic identity is

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}(X_i - \overline{X})^2 + n(\overline{X} - \mu)^2.$$

To see why this is true, write each deviation as

$$X_i - \mu = (X_i - \overline{X}) + (\overline{X} - \mu)$$

and square it. When we add over observations, the cross term is zero because the residuals add to zero.

The identity separates total squared departure from the population mean into two pieces. One describes spread around the sample mean. The other describes how far the sample mean is from the population mean.

9.6.1 How the normal density separates

For fixed population parameters, the joint density of the observations is proportional to

$$\exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right].$$

Substituting the identity above turns this into a product:

$$\exp\left[-\frac{n(\overline{x}-\mu)^2}{2\sigma^2}\right]\times\exp\left[-\frac{\sum_{i=1}^{n}(x_i-\overline{x})^2}{2\sigma^2}\right].$$

We can describe a sample by its mean and its first $n-1$ residuals. Those values determine the last residual and every original observation.

In these coordinates, the first factor depends only on the mean and the second only on the residuals. After normalization, they form separate probability densities. This factorization proves that the sample mean and the residual vector are independent. Since either estimator of the variance is calculated entirely from the residuals, it is independent of the sample mean as well.

9.6.2 Independence is a special normal-model result

The same conclusion may not hold for other populations. Consider two Bernoulli observations, each either 0 or 1. Their possible sample means and usual sample variances are

Observations	$\overline{X}$	S^2
0, 0	0	0
0, 1 or 1, 0	0.5	0.5
1, 1	1	0

Knowing the sample mean tells us the sample variance exactly. For a success probability strictly between 0 and 1, the two summaries are dependent.

9.7 Example: checking both the center and spread of cable lengths

A cutting machine produces independent lengths modeled as normal. For a sampling calculation, suppose the true mean is 120 centimeters and the true standard deviation is 1.5 centimeters. A technician checks a random batch of 12 pieces.

The batch meets two descriptive requirements if its sample mean is between 119.4 and 120.6 centimeters and its sample standard deviation is no greater than 1.8 centimeters. What proportion of batches meet both requirements under this model?

First, the sample mean has distribution

$$\overline{X} \sim N\left(120, \frac{1.5^2}{12}\right).$$

Thus,

$$P(119.4 \le \overline{X} \le 120.6) \approx 0.834.$$

For the spread requirement, $S \le 1.8$ is equivalent to $S^2 \le 1.8^2$. Since

$$\frac{11S^2}{1.5^2} \sim \chi^2_{11},$$

we obtain

$$P(S \le 1.8) = P\left(\frac{11S^2}{1.5^2} \le \frac{11(1.8)^2}{1.5^2}\right) \approx 0.853.$$

Independence lets us multiply the probabilities:

$$P(119.4 \le \overline{X} \le 120.6,\ S \le 1.8) \approx 0.834(0.853) \approx 0.711.$$

About 71.1% of batches would meet both requirements. The comma inside the probability means that both conditions must hold.

```
mean_probability <- pnorm(120.6, mean = 120, sd = 1.5 / sqrt(12)) -
  pnorm(119.4, mean = 120, sd = 1.5 / sqrt(12))
spread_probability <- pchisq(11 * 1.8^2 / 1.5^2, df = 11)
mean_probability * spread_probability
```

```
[1] 0.7113778
```

Among batches satisfying the spread requirement, the chance of satisfying the mean requirement is still about 0.834. Selecting batches with small sample spread does not make their sample means more accurate under this fixed normal model.

9.7.1 A simulation that separates batches by their observed spread

The following simulation asks whether batches with smaller observed standard deviations have a different distribution of sample means. It generates 10,000 batches, then separates them according to whether their sample standard deviation is at most 1.5 centimeters or greater than 1.5 centimeters.

```
set.seed(132)
batch_means <- numeric(10000)
batch_sds <- numeric(10000)

for (i in seq_along(batch_means)) {
  lengths <- rnorm(12, mean = 120, sd = 1.5)
  batch_means[i] <- mean(lengths)
  batch_sds[i] <- sd(lengths)
}

bin_edges <- seq(116, 124, by = 0.1)
possible_means <- seq(118, 122, length.out = 400)

par(mfrow = c(1, 2))

hist(
  batch_means[batch_sds <= 1.5],
  breaks = bin_edges, probability = TRUE,
  col = "skyblue", border = "white",
  xlim = c(118, 122), ylim = c(0, 1.1),
  xlab = "Sample mean (centimeters)", main = "Sample SD at most 1.5"
)
lines(
  possible_means, dnorm(possible_means, mean = 120, sd = 1.5 / sqrt(12)),
  col = "steelblue", lwd = 2
)

hist(
```

```r
  batch_means[batch_sds > 1.5],
  breaks = bin_edges, probability = TRUE,
  col = "skyblue", border = "white",
  xlim = c(118, 122), ylim = c(0, 1.1),
  xlab = "Sample mean (centimeters)", main = "Sample SD above 1.5"
)
lines(
  possible_means, dnorm(possible_means, mean = 120, sd = 1.5 / sqrt(12)),
  col = "steelblue", lwd = 2
)

par(mfrow = c(1, 1))
```

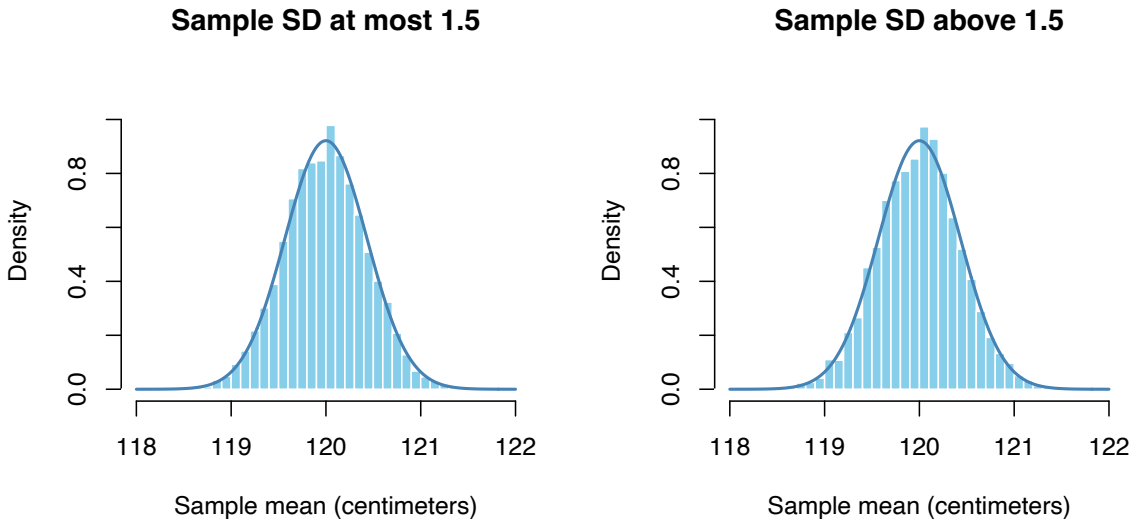


Figure 9.1: Sample means from batches selected by their observed spread. Both groups follow the same normal sampling density, shown in blue. Differences between the histograms reflect simulation variability.

The similar distributions in Figure 9.1 illustrate what independence means. Conditioning on one of these spread groups leaves the sampling distribution of the mean unchanged. The simulation supports the theoretical result, but a picture alone would not prove independence.

9.8 Looking ahead: estimating the standard error

The sample mean has true standard error $\sigma/\sqrt{n}$. When σ is unknown, a natural estimate is $S/\sqrt{n}$.

Replacing the fixed population standard deviation by the random sample standard deviation changes the distribution of a standardized mean. Chapter 10 will use the independence result from this chapter to work out that distribution exactly.

9.9 Chapter summary

- For independent normal observations, the sample mean is exactly normal with mean μ and variance σ^2/n.
- The MLE of variance divides the squared-deviation sum by n. The usual sample variance S^2 divides it by $n-1$.
- Estimating the mean forces the residuals to add to zero, leaving $n-1$ degrees of freedom.
- Both $n\hat{\sigma}^2/\sigma^2$ and $(n-1)S^2/\sigma^2$ have a chi-square distribution with $n-1$ degrees of freedom.
- The variance MLE is biased downward, while S^2 is unbiased.
- Under normal sampling, the sample mean and either sample variance are independent.
- Independence allows joint probabilities about sample location and spread to be calculated by multiplication.
- These exact results depend on the normal model. They do not hold for every population distribution.

10 The t Distribution

Suppose you estimate an average from a small sample. You know that sample means vary, so you want to express your estimation error in standard error units. There is one complication: the population standard deviation is unknown. You have to estimate it from the same observations.

Some samples will make that standard deviation look too large. Others will make it look too small. When the estimated standard error is small, even an ordinary difference between the sample mean and population mean can look large.

The **t distribution** accounts for this extra uncertainty. It looks much like a standard normal distribution near the center, but it allows more probability far from zero. This chapter explains where it comes from and why it appears when a normal population's standard deviation is estimated from data.

10.1 A normal variable divided by an estimated scale

Chapter 9 gave us two independent quantities from a normal sample: the sample mean and the sample variance. To see how they combine, we first define a distribution using a simpler pair of random variables.

> **Note**
>
> Z denotes a standard normal random variable, and Y denotes a chi-square variable with m degrees of freedom. We assume that Z and Y are independent.
> $T \sim t_m$ means that the random variable T has a t distribution with m degrees of freedom. The lowercase t names the distribution, while its subscript gives the degrees of freedom. It is also called Student's t distribution.

If

$$Z \sim N(0,1), \qquad Y \sim \chi^2_m,$$

independently, then

$$T = \frac{Z}{\sqrt{Y/m}} \sim t_m.$$

The numerator is standard normal. The denominator is positive and random. Since $E[Y] = m$, the quantity Y/m is centered at 1. But in any one realization of the random variable Y, it can be smaller or larger than 1.

When Y/m is small, division by its square root makes the magnitude of T larger. This is why the t distribution has heavier tails than a standard normal distribution. A large ratio can result from a large numerator, a small denominator, or both.

Independence is an important part of the definition. A ratio involving a normal variable and a chi-square variable does not automatically have a t distribution if those variables are dependent.

10.2 Shape, center, and spread

The t distribution is symmetric about zero. A positive numerator and an equally large negative numerator are equally likely, and the denominator is always positive. Thus positive and negative values of the same magnitude receive the same density.

The degrees of freedom control how heavy the tails are. With few degrees of freedom, large magnitudes are more common. As the degrees of freedom increase, the t distribution becomes closer to the standard normal distribution.

10.2.1 Why the normal distribution appears as a limit

For $Y \sim \chi^2_m$,

$$E\left[\frac{Y}{m}\right] = 1, \qquad \operatorname{Var}\left(\frac{Y}{m}\right) = \frac{2}{m}.$$

As m grows, Y/m becomes concentrated near 1. The denominator $\sqrt{Y/m}$ is then usually close to 1, so the ratio behaves increasingly like its standard normal numerator.

This explains why a normal approximation to a t distribution improves as the degrees of freedom grow. There is no sudden switch at a particular number of observations. The change is gradual.

10.2.2 A symmetric distribution may not have a mean

For $m > 1$, the t distribution has mean zero. For $m > 2$, its variance is

$$\text{Var}(T) = \frac{m}{m-2}.$$

This variance is greater than 1 for every finite $m > 2$, and it approaches 1 as m grows.

With very few degrees of freedom, the tails are so heavy that some expectations do not exist as finite values. Symmetry alone does not guarantee a mean of zero.

Degrees of freedom	Mean	Variance
$0 < m \leq 1$	Does not exist	Not defined because the mean does not exist
$1 < m \leq 2$	0	Infinite
$m > 2$	0	$m/(m-2)$

> **Note**
>
> For $k > 0$, the expression $E[\lvert T\rvert^k]$ is the expected value of the kth power of the magnitude of T. It is called an absolute moment. The paired bars mean absolute value, and k is a positive exponent.
> The statement $E[\lvert T\rvert^k] < \infty$ means this expectation is finite. The symbol ∞ denotes infinity.

For a t distribution, $E[\lvert T\rvert^k]$ is finite exactly when $k < m$. For example, the absolute first moment requires more than one degree of freedom, and the second moment requires more than two. This is a precise way to describe how the tails change with m.

10.3 The probability density

> **Note**
>
> $f_T(t)$ denotes the density of the random variable T at a possible numerical value t. The lowercase t inside the formula is the location where we evaluate the density.
> Γ is the gamma function. The symbol π in the constant below is the mathematical constant, rather than a prior distribution.

The density is

$$f_T(t) = \frac{\Gamma\left((m+1)/2\right)}{\sqrt{m\pi}\,\Gamma(m/2)} \left(1 + \frac{t^2}{m}\right)^{-(m+1)/2}, \qquad -\infty < t < \infty.$$

The first factor makes the total area equal to 1. The second determines the shape. Because t appears through t^2, the density is the same at t and $-t$.

We usually calculate t probabilities with a table or software rather than integrating this formula by hand. Still, the density helps explain why a t curve has a familiar bell shape with more probability in its tails.

```
center_values <- seq(-4, 4, length.out = 600)
tail_values <- seq(2, 6, length.out = 400)

par(mfrow = c(1, 2))

plot(
  center_values, dnorm(center_values),
  type = "l", col = "gray40", lwd = 2,
  xlab = "Value", ylab = "Density", main = "Center and tails"
)
lines(center_values, dt(center_values, df = 2), col = "darkorange", lwd = 2, lty = 2)
lines(center_values, dt(center_values, df = 8), col = "steelblue", lwd = 2, lty = 3)
legend(
  "topright", legend = c("Normal", "t, df = 2", "t, df = 8"),
  col = c("gray40", "darkorange", "steelblue"),
  lwd = 2, lty = c(1, 2, 3), bty = "n"
)

plot(
  tail_values, dnorm(tail_values),
  type = "l", col = "gray40", lwd = 2,
  ylim = c(0, 0.075),
  xlab = "Value", ylab = "Density", main = "A closer look at the tail"
)
lines(tail_values, dt(tail_values, df = 2), col = "darkorange", lwd = 2, lty = 2)
lines(tail_values, dt(tail_values, df = 8), col = "steelblue", lwd = 2, lty = 3)

par(mfrow = c(1, 1))
```

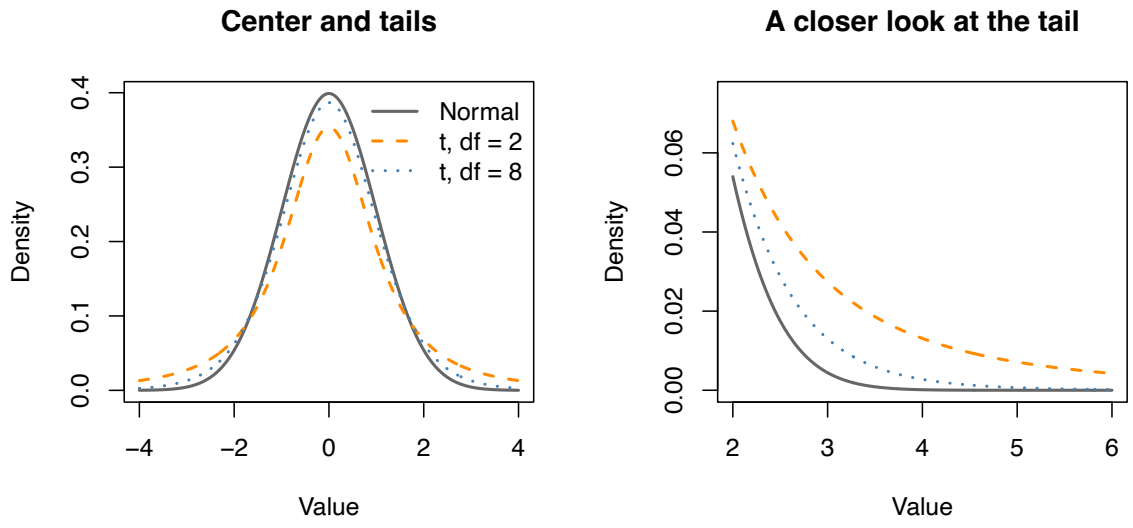


Figure 10.1: The t and standard normal densities have similar centers, but the t densities have more probability in the tails. The right panel shows the positive tail in more detail.

The right panel in Figure 10.1 makes a difference visible that is easy to miss near the center. Using a normal curve in place of a t curve can substantially understate the probability of a large standardized error when the degrees of freedom are small.

10.4 How the t distribution arises from a normal sample

Suppose $X_1, \ldots, X_n$ are independent normal observations with unknown mean μ and unknown variance σ^2, with $n \geq 2$. Recall that

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

If the population standard deviation were known, we could standardize the sample mean using

$$Z = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1).$$

Chapter 9 also tells us that

$$Y = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

The sample mean and sample variance are independent, so Z and Y are independent. These are exactly the ingredients in the definition of a t variable, with $m = n - 1$.

Substituting them into the ratio gives

$$\begin{aligned}\frac{Z}{\sqrt{Y/(n-1)}} &= \frac{(\overline{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{S^2/\sigma^2}} \\ &= \frac{\overline{X} - \mu}{S/\sqrt{n}}.\end{aligned}$$

The unknown population standard deviation cancels. Therefore,

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} = \frac{\sqrt{n}(\overline{X} - \mu)}{S} \sim t_{n-1}.$$

> **Note**
>
> $S/\sqrt{n}$ estimates the standard error of the sample mean. Dividing $\overline{X} - \mu$ by it expresses the estimation error in estimated standard error units.
> S is the square root of the sample variance with denominator $n - 1$. It is not the square root of the variance MLE. The degrees of freedom are $n-1$ because the chi-square variable in the denominator has $n - 1$ degrees of freedom.

For example, a value of $T = 2$ means the sample mean is two estimated standard errors above the population mean. A value of $T = -2$ means it is two estimated standard errors below.

The sample mean itself is still normally distributed. It is this ratio, with its random denominator, that has a t distribution. The exact result uses independent normal observations. A small sample by itself does not justify a t model for arbitrary data.

10.5 Why the choice of cutoff matters

A standard normal variable falls between approximately -1.96 and 1.96 with probability 0.95. If we divide an error by an estimated standard error, those same cutoffs generally capture less than 95% of the probability.

The following figure shows the exact probability

$$P(-1.96 \leq T \leq 1.96), \qquad T \sim t_{n-1},$$

using the more precise standard normal cutoff in the code.

```
sample_sizes <- 3:60
normal_cutoff <- qnorm(0.975)
captured_probability <- 2 * pt(normal_cutoff, df = sample_sizes - 1) - 1

plot(
  sample_sizes, captured_probability,
  type = "l", col = "steelblue", lwd = 2,
  ylim = c(0.78, 0.96),
  xlab = "Sample size", ylab = "Probability between the cutoffs",
  main = "Using a normal cutoff with an estimated standard error"
)
abline(h = 0.95, col = "firebrick", lty = 2, lwd = 2)
```

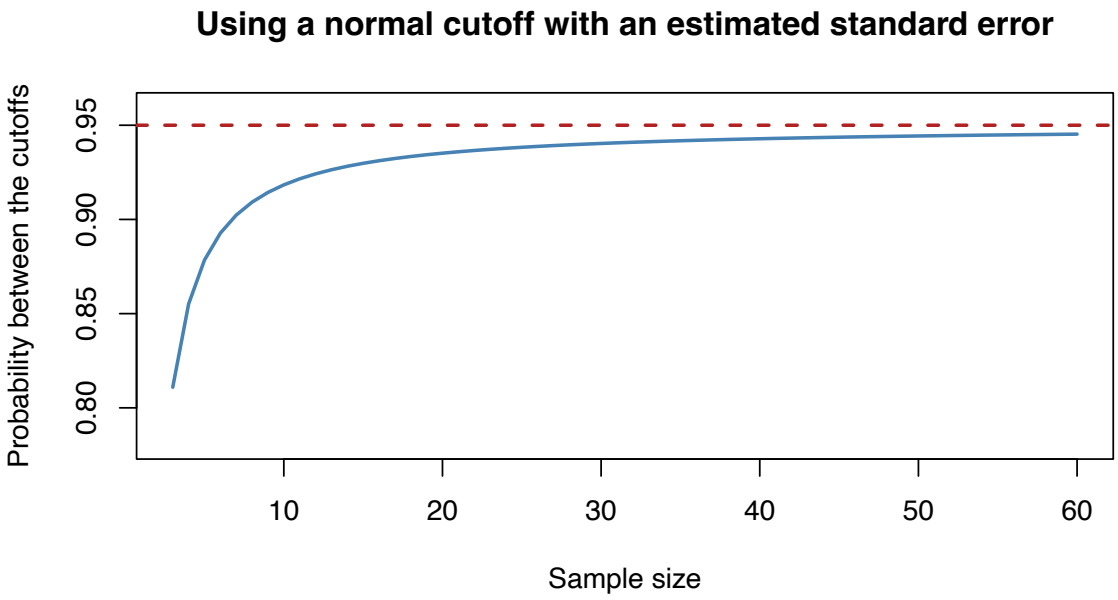


Figure 10.2: The usual 95% normal cutoff captures less than 95% of a t distribution. The difference becomes smaller as sample size increases.

The curve in Figure 10.2 is below the desired probability, especially for small samples. To

capture the same central probability with a t distribution, we need cutoffs farther from zero.

10.6 Reading a t table and finding quantiles

> **Note**
>
> $t_p(m)$ denotes the cutoff with cumulative probability p under a t distribution with m degrees of freedom:
>
> $$P(T \leq t_p(m)) = p.$$
>
> The subscript p specifies probability to the left of the cutoff, and the argument m specifies degrees of freedom. This notation gives a numerical cutoff, while T denotes a random variable.

The table in Lecture 10 uses this left-tail convention. For a central interval containing 90% of a symmetric distribution, 5% remains in each tail. Its positive endpoint is therefore the 0.95 quantile, not the 0.90 quantile.

For 15 degrees of freedom, the table gives

$$t_{0.95}(15) \approx 1.753.$$

By symmetry,

$$P(-1.753 \leq T \leq 1.753) \approx 0.90, \qquad T \sim t_{15}.$$

The same cutoff can be calculated with `qt(0.95, df = 15)`. More generally, symmetry gives $t_{1-p}(m) = -t_p(m)$.

R function	Purpose
`dt(x, df = m)`	Evaluate a t density at `x`
`pt(x, df = m)`	Calculate probability at or below `x`
`pt(x, df = m, lower.tail = FALSE)`	Calculate probability above `x`
`qt(p, df = m)`	Find the cutoff with probability `p` to its left
`rt(n, df = m)`	Generate `n` random t values with `m` degrees of freedom

For a positive cutoff, the probability beyond that cutoff in either direction is twice the upper-tail probability. The two tails are equal because the distribution is symmetric.

10.7 Example: comparing two timing methods on the same recordings

An audio lab uses an automated method and a manual method to mark the start of a sound. Each recording is measured both ways. Different recordings may have very different start times, so the laboratory compares the two methods within each recording.

> **Note**
>
> D_i is the automated time minus the manual time for recording i. A positive difference means the automated method places the start later.
> μ_D and σ_D^2 are the population mean and variance of these differences. $\overline{D}$ is the sample mean difference, and S_D is the sample standard deviation of the differences, calculated with divisor $n - 1$.
> The subscript D identifies quantities describing the differences. Each recording supplies one difference, so n counts recordings rather than individual timing measurements.

Suppose the differences across recordings are independent and approximately normal. Eight recordings give these differences, in milliseconds:

$$1.4, \quad 0.8, \quad 1.1, \quad -0.1, \quad 0.6, \quad 1.5, \quad 0.3, \quad 0.8.$$

The sample summaries are

$$\overline{d} = 0.8, \qquad s_D \approx 0.540, \qquad \frac{s_D}{\sqrt{8}} \approx 0.191.$$

The lab wants to compare this observed difference with the model in which the methods have no systematic offset, so $\mu_D = 0$. Under an exact normal model with that mean,

$$\frac{\overline{D} - 0}{S_D/\sqrt{8}} \sim t_7.$$

The observed value is

$$\frac{0.8}{0.540/\sqrt{8}} \approx 4.19.$$

The estimated mean difference is about 4.19 estimated standard errors above zero.

```
differences <- c(1.4, 0.8, 1.1, -0.1, 0.6, 1.5, 0.3, 0.8)
mean_difference <- mean(differences)
```

```r
standard_error <- sd(differences) / sqrt(8)
observed_t <- mean_difference / standard_error
observed_t
```

```
[1] 4.191495
```

```r
2 * pt(abs(observed_t), df = 7, lower.tail = FALSE)
```

```
[1] 0.004078467
```

The probability of a ratio at least this far from zero in either direction is about 0.0041 under the zero-offset normal model. This tells us that the observed standardized difference would be unusual under that model. It is not the probability that the population mean difference equals zero.

Note that the two measurements on a single recording could be dependent. However, we analyze one difference per recording and only require independence across those differences. We also require the normal model to describe the differences well for the exact t result to apply.

10.8 Example: allowing for one future processing delay

A technician records processing delays from 16 independent runs of a device. Suppose the delays follow a normal distribution with unknown mean and variance. The observed sample mean is 42 milliseconds and the sample standard deviation is 4 milliseconds.

The technician wants a range for the delay on the next run. Predicting a single future observation requires allowing for its own variability as well as the uncertainty in the estimated mean.

> **Note**
>
> X_{new} denotes a future observation from the same population, independent of the current sample. The text subscript "new" is a label.
> $X_{\text{new}} - \overline{X}$ is the error made when predicting that observation with the current sample mean. Its standard deviation is $\sigma\sqrt{1 + 1/n}$. We estimate this by $S\sqrt{1 + 1/n}$.

Because the new observation and the sample mean are independent normal quantities,

$$X_{\text{new}} - \overline{X} \sim N\left(0, \sigma^2\left(1 + \frac{1}{n}\right)\right).$$

The variance is a sum. The term σ^2 comes from the new observation and the term σ^2/n comes from the sample mean.

The sample mean is independent of S, and the new observation is independent of the entire current sample. Thus the prediction error is independent of S. Dividing its standardized normal version by the same chi-square scale as before gives

$$\frac{X_{\text{new}} - \overline{X}}{S\sqrt{1 + 1/n}} \sim t_{n-1}.$$

For $n = 16$, the central 90% cutoff is $t_{0.95}(15) \approx 1.753$. Rearranging the probability statement gives

$$P\left(\overline{X} - 1.753S\sqrt{1 + 1/16} \leq X_{\text{new}} \leq \overline{X} + 1.753S\sqrt{1 + 1/16}\right) \approx 0.90.$$

Using the observed sample summaries, the endpoints are

$$42 - 1.753(4)\sqrt{1 + 1/16} \approx 34.77,$$

$$42 + 1.753(4)\sqrt{1 + 1/16} \approx 49.23.$$

```
prediction_margin <- qt(0.95, df = 15) * 4 * sqrt(1 + 1 / 16)
c(42 - prediction_margin, 42 + prediction_margin)
```

```
[1] 34.77199 49.22801
```

This gives a prediction interval from about 34.77 to 49.23 milliseconds. Under repeated sampling and future observations from the normal model, this construction contains the future delay 90% of the time.

The margin is about 7.23 milliseconds. Using the mean's estimated standard error, $4/\sqrt{16} = 1$, would give a much smaller margin of about 1.75 milliseconds with the same t cutoff. That smaller scale describes uncertainty about the population mean. It leaves out the variation of the next individual delay.

Even with a very large sample, a single future observation still varies around the population mean. More observations help us locate that mean, but they do not remove individual variation.

10.9 Chapter summary

- A t variable is a standard normal variable divided by the square root of an independent chi-square variable divided by its degrees of freedom.
- The t distribution is symmetric about zero and has heavier tails than the standard normal distribution.
- Its mean exists for more than one degree of freedom. Its variance is finite only for more than two degrees of freedom.
- As the degrees of freedom grow, the t distribution approaches the standard normal distribution.
- For independent normal observations, $(\overline{X} - \mu)/(S/\sqrt{n})$ has a t distribution with $n - 1$ degrees of freedom.
- The denominator $n-1$ in S^2 matters. Substituting the variance MLE without adjusting the formula changes the statistic's distribution.
- A central probability uses two tails. A central 90% interval requires the positive 0.95 quantile.